Prompt Engineering from Zero

A beginner's notebook · 15 pages · no coding needed

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Framework inspired by Google's Prompting Essentials: Task, Context, References, Evaluate, Iterate.

Rule of the class:
we learn AI by **using** it.

1. What is a prompt?

A prompt is the instruction you give an AI. Think of the AI as a very fast intern: widely read, eager, but it knows nothing about *your* situation until you say so.

Weak prompt

"Write about dogs."

Which dogs? For whom? How long? The AI has to guess.

Strong prompt

"Write a friendly 150-word Instagram caption for a rescue shelter, with 3 reasons to adopt."

> **Try it:** run both in any AI tool and compare the answers.

2. How AI reads your prompt

- It predicts the most likely next words from patterns it learned.
- It sees only what you type plus the chat so far.
- It can sound confident and still be wrong.
- Clearer input gives better output.

You → Prompt → AI model → Output → **You review**

The last step is the one people skip. **You** are always the editor.

3. The 5-step framework: T C R E I

Memory trick: **T**houghtfully **C**reate **R**eally **E**xcellent **I**nputs

Ⓣ **Task**: say exactly what you want done.

Ⓒ **Context**: who it is for, why, and the limits.

Ⓡ **References**: show an example of what good looks like.

Ⓔ **Evaluate**: check the answer before you use it.

Ⓘ **Iterate**: refine with follow-ups instead of starting over.

T is for Task

Start with a strong verb. Add a **persona** (who the AI should act as) and a **format** (what the answer looks like).

[Persona] + [Action] + [Format]

> *"You are a career coach. Write a LinkedIn headline for a fresher in data analytics. Give 5 options in a numbered list, under 12 words each."*

- Good verbs: write, summarise, compare, explain, plan, rewrite.
- One clear job per prompt works best.

> **Try it:** turn "help with my resume" into a task with persona and format.

C is for Context

Context is everything the AI cannot guess. Cover these four:

- **Audience**: college freshers, busy managers
- **Goal**: why you need this
- **Constraints**: length, deadline, budget
- **Tone**: warm, formal, playful

"Explain what an API is. Audience: first-year B.Com students with no tech background. Goal: a 2-minute intro for a workshop. Tone: friendly, with one everyday analogy."

> **Try it:** ask for the same explanation for a child, then for a CEO. Notice the shift.

R is for References

Show, don't just tell. One or two examples teach style faster than a paragraph of rules. (Pros call this *few-shot prompting*.)

> *"Here are two taglines I like:*
> *1. 'Fuel your day, one sip at a time.'*
> *2. 'Brewed slow. Enjoyed fast.'*
> *Write 3 new taglines for a mountain homestay in the same style."*

- Examples can be text, a table, or a format you want copied.
- Never paste private or confidential material into a public AI tool.

E is for Evaluate

AI can invent facts, numbers, and links (this is called a hallucination). Check every output with this list:

- ☐ Accurate: are facts verified?
- ☐ Relevant: does it answer my task?
- ☐ Complete: is anything missing?
- ☐ Tone: does it sound like me?
- ☐ Safe: nothing private or biased?
- ☐ Useful: can I use it as is?

> **Try it:** ask an AI for "3 statistics about remote work", then verify each one with a trusted source.

Trust,
then **verify**.

I is for Iterate

Your first prompt is a draft. Keep the chat going and steer it, like giving feedback to a colleague.

Prompt → Output → Evaluate → Refine ↺

If the answer is…	Say…
Too long	*"Cut this to 100 words."*
Too complex	*"Explain it like I'm 15."*
Too generic	*"Add a real example from retail."*
Not enough	*"Give 3 more options, different tone."*
Unsure what is missing	*"What would make this stronger?"*

Putting it together

Before

"Make a workshop plan on AI."

After (T C R E I)

Task: You are a corporate trainer. Create a 60-minute workshop plan on prompt engineering.

Context: 30 college freshers, no tech background, goal is to write a good prompt by the end.

Reference: follow this layout: Time | Activity | Outcome.

Format: table, then 3 practice exercises.

Then **evaluate** the plan and **iterate**: "Add a 5-minute icebreaker."

4 techniques worth knowing

Technique	What it does	Example
Role prompting	Sets expertise and viewpoint	*"Act as a financial analyst…"*
Few-shot	Teaches style with examples	*"Here are 2 samples. Write a third."*
Step by step	Improves reasoning and maths	*"Solve this and show each step."*
Format control	Shapes the structure	*"Answer as a table with 3 columns."*

> **Try it:** combine two techniques in one prompt.

One framework, any tool

ChatGPT
General-purpose assistant. Good for drafting, brainstorming, and quick Q&A.

Claude
Strong at long documents, careful writing, and following detailed instructions.

Gemini
Works closely with Google apps like Docs, Sheets, and Gmail.

Tools update often, but the framework stays the same.

> **Try it:** send one T C R E I prompt to all three. Compare accuracy, tone, and structure, then note which suited the task.

Everyday uses

Area	Sample prompt
Excel / Sheets	*"Write a formula to sum column C where column A is 'North', and explain it simply."*
Email	*"Rewrite this email to be polite but firm. Keep it under 80 words."*
Content	*"Give 10 reel ideas for a budget travel channel, with a hook for each."*
Learning	*"Make a 7-day plan to learn pivot tables, 30 minutes a day."*
Summaries	*"Summarise this report in 5 bullets and list 2 risks."*

Common mistakes and fixes

Mistake	Fix
Too vague	Add audience, goal, and length.
Five tasks in one prompt	Split into steps; one job at a time.
No format asked	Say table, bullets, or word count.
Trusting blindly	Verify facts, figures, and links.
Sharing private data	Remove names, numbers, and company secrets first.
Giving up after one try	Iterate at least twice.

Your turn: 15-minute exercise

1. Pick one real task from your week.
2. Write a prompt using **T C R E I**.
3. Run it, then **evaluate** with the checklist.
4. **Iterate** twice and save your best version.

Cheat sheet

Task = verb + persona + format · **Context** = audience, goal, limits, tone · **Reference** = 1 to 2 examples · **Evaluate** = verify facts · **Iterate** = refine, don't restart

Next: AI in Excel, content creation, and workflow automation. **Keep practising!**